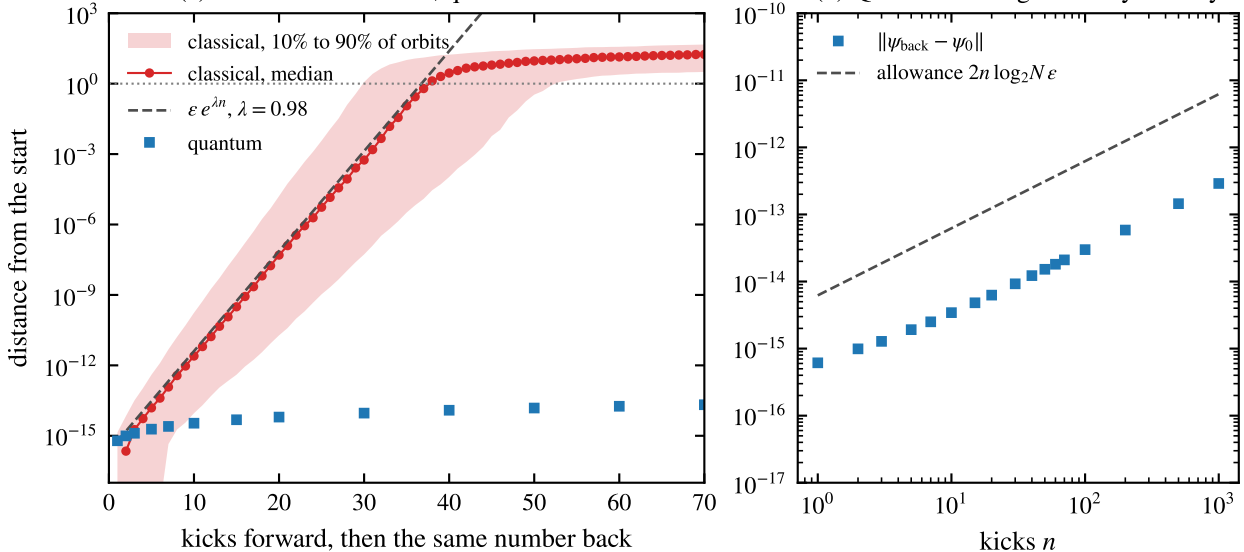
(a) Classical return fails, quantum does not
classical, 10% to 90% of orbits
classical, median
$\varepsilon\, e^{\lambda n}$, $\lambda = 0.98$
quantum
distance from the start
kicks forward, then the same number back
(b) Quantum error grows only linearly
$\|\psi_{\text{back}} - \psi_0\|$
allowance $2n \log_2 N \varepsilon$
kicks n